

Solutions: Problem Set 7

Poisons, Rods, and Flux Shaping

CBE 30235

Problem 1: Equilibrium Xenon

(a) Burnout vs. Decay

We compare the removal rate constants for Xenon-135: radioactive decay (λ_X) and neutron absorption ("burnout", $\sigma_a^X \phi$).

Given:

- $\lambda_X = 2.1 \times 10^{-5} \text{ s}^{-1}$
- $\sigma_a^X = 2.6 \times 10^6 \text{ barns} = 2.6 \times 10^{-18} \text{ cm}^2$
- $\phi = 10^{13} \text{ n/cm}^2\text{s}$

Calculating the burnout rate:

$$\sigma_a^X \phi = (2.6 \times 10^{-18} \text{ cm}^2)(10^{13} \text{ s}^{-1}) = 2.6 \times 10^{-5} \text{ s}^{-1}$$

Conclusion: Since $2.6 \times 10^{-5} > 2.1 \times 10^{-5}$, the **Burnout mechanism dominates**. At this flux level, more Xenon atoms are destroyed by neutron capture than by radioactive decay.

(b) Equilibrium Concentration Change

The equilibrium concentration is proportional to:

$$X_\infty \propto \frac{\phi}{\lambda_X + \sigma_a^X \phi}$$

Case 1 ($\phi = 10^{13}$):

$$\text{Denom}_1 = 2.1 \times 10^{-5} + 2.6 \times 10^{-5} = 4.7 \times 10^{-5}$$

$$X_1 \propto \frac{1}{4.7} \approx 0.213$$

Case 2 ($\phi = 2 \times 10^{13}$):

$$\sigma_a^X \phi_{\text{new}} = 5.2 \times 10^{-5}$$

$$\text{Denom}_2 = 2.1 \times 10^{-5} + 5.2 \times 10^{-5} = 7.3 \times 10^{-5}$$

$$X_2 \propto \frac{2}{7.3} \approx 0.274$$

Ratio:

$$\frac{X_2}{X_1} = \frac{0.274}{0.213} \approx \mathbf{1.28}$$

Even though power doubled, Xenon concentration only increased by **28%** because of the "saturation" effect (increased burnout).

(c) Reactivity Worth

First, we find the ratio of the macroscopic absorption of Xenon to the macroscopic fission cross-section using the formula:

$$\Sigma_a^X = \sigma_a^X X_\infty = \frac{\gamma \Sigma_f \sigma_a^X \phi}{\lambda_X + \sigma_a^X \phi}$$

Rearranging this to solve for Σ_a^X / Σ_f and using the given yield $\gamma = 0.06$:

$$\frac{\Sigma_a^X}{\Sigma_f} = \frac{\gamma \sigma_a^X \phi}{\lambda_X + \sigma_a^X \phi} = \frac{0.06(2.6 \times 10^{-5})}{4.7 \times 10^{-5}} = \frac{0.156}{4.7} \approx 0.03319$$

Next, we calculate the negative reactivity worth using the given approximation and the ratio $\Sigma_f / \Sigma_a^{Total} = 1/1.2$:

$$\begin{aligned} \rho_{Xe} &\approx -\frac{\Sigma_a^X}{\Sigma_a^{Total}} = -\left(\frac{\Sigma_a^X}{\Sigma_f}\right) \left(\frac{\Sigma_f}{\Sigma_a^{Total}}\right) \\ \rho_{Xe} &\approx -(0.03319) \left(\frac{1}{1.2}\right) \approx -0.02766 \end{aligned}$$

Answer: The equilibrium Xenon worth is approximately **-2.77% $\Delta k/k$** (or **-2766 pcm**).

Problem 2: The Iodine Pit

(a) Physical Explanation

Immediately after shutdown ($\phi \rightarrow 0$):

1. **Production Continues:** The Iodine-135 accumulated during operation continues to decay into Xenon-135 ($t_{1/2} = 6.6$ hr).
2. **Removal Stops:** The primary removal mechanism (burnout, $\sigma_a^X \phi$) vanishes instantly.
3. **Result:** Since production > decay removal, the Xenon concentration rises sharply.

(b) Operational Implication

The "Iodine Pit" creates a temporary window where the reactor cannot be restarted.

- If the Xenon peak worth (-8%) exceeds the available shutdown margin/excess reactivity (e.g., $+5\%$), the reactor becomes **subcritical** regardless of control rod position.
- Operators have a short window (30–60 mins) to restart before the Xenon builds up too high. If they miss this, the reactor is "poisoned out" for roughly 2 days until the Xenon naturally decays.

(c) 50% vs 100% Power

The equilibrium Iodine inventory is directly proportional to the fission rate (Power). A reactor scrambling from 100% power has roughly ****twice**** the stored Iodine as one at 50% power. Consequently, the resulting Xenon peak will be significantly higher, making the "dead time" harder to manage.

Problem 3: The S-Curve

Equation: $\rho(z) = \rho_{total} \left[\frac{z}{H} - \frac{1}{2\pi} \sin\left(\frac{2\pi z}{H}\right) \right]$ with $\rho_{total} = \$2.00$.

(a) Bottom 25% ($z = 0$ to 100 cm)

At $z = 100$, $z/H = 0.25$.

$$\rho(100) = 2.00 \left[0.25 - \frac{1}{2\pi} \sin\left(\frac{\pi}{2}\right) \right] = 2.00[0.25 - 0.159] = 2.00(0.091)$$

$$\rho(0 \rightarrow 100) = \mathbf{\$0.18}$$

(b) Middle 25% ($z = 150$ to 250 cm)

At $z = 150$, $z/H = 0.375$.

$$\rho(150) = 2.00 \left[0.375 - \frac{\sin(135^\circ)}{2\pi} \right] = 2.00[0.375 - 0.113] = \$0.524$$

At $z = 250$, $z/H = 0.625$.

$$\rho(250) = 2.00 \left[0.625 - \frac{\sin(225^\circ)}{2\pi} \right] = 2.00[0.625 - (-0.113)] = \$1.476$$

$$\Delta\rho = \$1.476 - \$0.524 = \mathbf{\$0.95}$$

(c) Comparison

The middle section inserts roughly **5 times more reactivity** than the bottom section (\$0.95 vs \$0.18). **Reason:** Rod worth is proportional to ϕ^2 . The flux is peaked in the center of the core and near zero at the bottom, so rod movement in the center affects far more neutrons.

Problem 4: Flux Shaping (F_q)**(a) Average Power Density**

$$P_{avg} = \frac{P_{total}}{V} = \frac{3000 \text{ MW}}{30 \text{ m}^3} = \mathbf{100 \text{ MW/m}^3}$$

(b) Peak Power Density

Using the given natural peaking factor $F_q = 3.64$:

$$P_{max} = P_{avg} \times F_q = 100 \times 3.64 = \mathbf{364 \text{ MW/m}^3}$$

(c) Zone Loading Benefit

New $F_q = 2.2$. Limit $P_{max} \leq 350 \text{ MW/m}^3$. Calculate the allowed average power density:

$$P_{avg}^{new} = \frac{350}{2.2} \approx 159.1 \text{ MW/m}^3$$

Calculate new Total Power:

$$P_{total}^{new} = 159.1 \text{ MW/m}^3 \times 30 \text{ m}^3 = \mathbf{4773 \text{ MW}}$$

Benefit: We can increase reactor power by **1773 MW** without melting the fuel, simply by flattening the flux profile.

Problem 5: Samarium vs. Xenon**(a) Shutdown Behavior**

Sm-149 is stable. It is produced by Pm-149 decay ($\tau \approx 53 \text{ hr}$). Since flux is zero, Sm-149 is not burned out, but it continues to be produced by the decaying Promethium. **Result:** Sm concentration **increases** to a constant saturation value and stays there indefinitely.

(b) Startup Behavior

Upon restart, the Sm concentration is higher than equilibrium. As the flux returns, the Sm is burned out, causing the concentration to **decrease** back to equilibrium.

(c) Operational Concern

Sm-149 is less dangerous than Xenon because:

1. It is stable; it does not disappear, so it is a fixed penalty rather than a time-dependent "window."
2. Its reactivity worth is generally smaller than the massive Xenon peak.